

ADVANCED DOCKER

ROADMAP

Step-by-Step Guide to Become a Docker Expert



LEARN

PRACTICE

BUILD

DEPLOY

MASTER

1



Deep Dive into Docker Architecture

- Docker Engine Architecture
- Namespaces, Cgroups, Union FS
- How Docker Daemon Works
- Docker Objects: Images, Containers, Layers

1–2 Weeks

Resources
Docker Docs
Awesome Docker

2



Advanced Dockerfile Techniques

- Multi-stage Builds
- Build Arguments & Environment Variables
- Best Practices for Smaller Images
- .dockerignore and Build Context

1–2 Weeks

Resources
Docker Docs
vsupalov.com

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Advanced Image Management

- Image Layers & Caching
- Image Optimization & Security Scanning
- Image Signing (Notary, Cosign)
- Private Registries

1–2 Weeks

Resources
Docker Docs
Notary Project

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Advanced Container Management

- Resource Limits (CPU, Memory, PIDs)
- Restart Policies, Healthchecks
- Logging Drivers
- Container Lifecycle & Cleanup

1–2 Weeks

Resources
Docker Docs

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Networking Deep Dive

- Bridge, Host, None, Overlay Networks
- Custom Networks & DNS
- Port Mapping & NAT
- Network Drivers (Macvlan, IPVLAN)

1–2 Weeks

Resources
Docker Docs
Linux Networking

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Storage & Volumes

- Bind Mounts vs Volumes
- Volume Drivers
- tmpfs Mounts
- Backup & Restore Volumes

1–2 Weeks

Resources
Docker Docs

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Security in Docker

- Docker Security Best Practices
- User Namespaces
- Capabilities & Seccomp
- Image Scanning (Trivy, Clair)

1–2 Weeks

Resources
Docker Docs
Anchore / Trivy

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Advanced Docker Compose

- Compose File v3+ Features
- Multiple Compose Files
- Profiles & Secrets
- Environment Management

1–2 Weeks

Resources
Compose Docs

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Docker Swarm (Orchestration)

- Swarm Architecture
- Services, Stacks, Secrets, Configs
- Rolling Updates & Rollbacks
- Scaling & Load Balancing

1–2 Weeks

Resources
Docker Docs

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CI/CD with Docker

- Optimized Docker Builds in CI
- Caching in CI/CD Pipelines
- Docker in Jenkins, GitHub Actions, GitLab CI
- Automated Testing with Containers

1–2 Weeks

Resources
Docs
CI/CD Guides

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Docker with Kubernetes

- Running Containers in K8s
- Images, Pods, Deployments
- ConfigMaps, Secrets, Volumes
- Helm Basics

2–3 Weeks

Resources
Kubernetes Docs
Helm Docs

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Real World Projects

- Build & Deploy Multi-tier Application
- CI/CD Pipeline with Docker
- Monitoring & Logging Setup
- Containerized Microservices

Ongoing

Resources
Personal Projects
Open Source

CORE SKILLS YOU WILL BUILD

- Containerization Deep Understanding
- High Performance & Optimized Images
- Secure & Reliable Containers
- Advanced Networking & Storage
- Scalable & Orchestrated Deployments
- Automation & CI/CD Integration
- Troubleshooting & Problem Solving

ADVANCED DOCKER FEATURES

- | | |
|-------------------------|-----------------------|
| Multi-stage Builds | Cgroups & Namespaces |
| BuildKit | User Namespaces |
| Secrets Management | Rootless Docker |
| Healthchecks | Live Restore |
| Logging Drivers | Experimental Features |
| Plugins & Extensibility | |

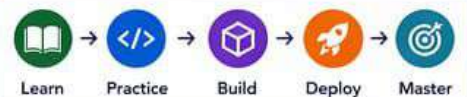
DOCKER ECOSYSTEM & TOOLS



BEST PRACTICES

- ✓ Use lightweight base images (Alpine, Distroless)
- ✓ Follow least privilege principle
- ✓ Keep images small & secure
- ✓ Scan images regularly
- ✓ Use .dockerignore
- ✓ Tag images properly
- ✓ Automate builds & deployments
- ✓ Monitor & clean up unused resources

LEARNING PATH FLOW



CAREER OPPORTUNITIES

- | | |
|-----------------------------|----------------------|
| • DevOps Engineer | • Platform Engineer |
| • Site Reliability Engineer | • Systems Engineer |
| • Cloud Engineer | • Container Engineer |



Consistency + Practice + Real Projects = Docker Mastery

